

Technology Stack (Architecture & Stack)

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Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Technical Architecture Summary

- **Application Logic / Technology Blocks:** Catalog UI Policies, Flow Designer Workflow, Subflows, Script Includes, and ATF Test Suites.
- **Infrastructural Demarcation:** ServiceNow SaaS Enterprise Cloud (Hosting Catalog & Flow Engine) integrated with On-Premise Network Infrastructure via a secure **MID Server Cluster**.
- **External Interfaces:** IntegrationHub REST / SSH Spokes connecting to Network Switches/Firewalls and SMTP Email Servers.
- **Data Storage:** ServiceNow Relational Cloud Database (sc_req_item, sc_task, sysapproval_approver) and CMDB IRE (cmdb_ci_netgear).
- **Machine Learning Interface:** ServiceNow Predictive Intelligence for automated request categorisation and priority prediction.

Technical Architecture:



Table-1 : Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	Interface where end users submit network requests and view real-time status updates.	ServiceNow Service Portal, HTML5, CSS3, AngularJS, Bootstrap
2.	Application Logic-1	Client-side form logic for dynamic field toggling and IPv4 Regex pattern validation.	ServiceNow JavaScript (Catalog Client Scripts, Catalog UI Policies)
3.	Application Logic-2	Server-side workflow orchestration, conditional approval routing, and catalog task generation.	ServiceNow Flow Designer, Subflows, Script Includes
4.	Application Logic-3	Security governance and record-level access control policies.	ServiceNow Access Control Lists (ACLs), Role-Based Access (RBAC)
5.	Database	Stores request items, fulfillment tasks, form variable values, and approval logs.	ServiceNow Relational Database (sc_req_item, sc_task, sysapproval_approver)
6.	Cloud Database	Configuration Item tracking datastore for network hardware and IP infrastructure.	ServiceNow CMDB (cmdb_ci_netgear, cmdb_ci_ip_address) via IRE
7.	File Storage	Storage repository for network topology diagrams, security specifications, and approval docs.	ServiceNow Attachment API / Database Storage (sys_attachment)

S.No	Component	Description	Technology
8.	External API-1	Integration layer to trigger automated switch/firewall configuration changes.	ServiceNow IntegrationHub (REST Spokes, SSH Command Spokes)
9.	External API-2	Outbound notification engine with dynamic email approval action buttons.	ServiceNow System Notification Engine (SMTP/POP3 API)
10.	Machine Learning Model	Automated request classification, routing prediction, and anomaly detection.	ServiceNow Predictive Intelligence (Classification / Similarity Engine)
11.	Infrastructure (Server / Cloud)	Multi-tenant SaaS cloud platform integrated with local network MID Server agents.	ServiceNow Enterprise Cloud Infrastructure, On-Prem MID Server Cluster

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Front-end portal libraries and scripting components leveraged within ServiceNow Service Portal.	AngularJS, Bootstrap, FontAwesome, JavaScript ES6
2.	Security Implementations	Data encryption at rest/in transit, role-based access rules, and secure MID Server communication.	AES-256 Encryption, HTTPS/TLS 1.3, ServiceNow RBAC ACLs, OAuth 2.0
3.	Scalable Architecture	Multi-instance SaaS platform supporting enterprise-wide concurrent request execution.	ServiceNow Advanced High Availability (AHA) Multi-Instance Architecture
4.	Availability	High availability infrastructure ensuring continuous workflow execution for critical network changes.	99.9% Cloud Uptime SLA, Redundant MID Server Clustering, Load Balancers
5.	Performance	Design considerations for rapid form loading, caching, and asynchronous workflow execution.	GlideRecord Indexing, Service Portal Caching, Asynchronous Flow Execution